Dazaet Galicia-Badillo

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10 October 2026

Dr Daniel Lee
Department of Chemical Engineering, School of Engineering
The University of Manchester
Manchester M13 9PL, United Kingdom

Re: Research Associate in Solid-State NMR Spectroscopy, ref. SAE-032399

Dear Dr Lee,

I am writing to apply for the position of Research Associate in Solid-State NMR Spectroscopy (ref. SAE-032399) within the EPSRC Programme Grant on Molecular Solid Solutions. I completed my PhD in Chemical Sciences at the Institute of Chemistry, UNAM (Mexico), in 2026 under the supervision of Prof. Braulio Rodríguez-Molina, and the degree was awarded with honours.

Solid-state NMR was the central technique of my PhD. On a Bruker 500 MHz spectrometer with a double-resonance (^{1}H/X) MAS probe, I set up my own experiments, including tuning and matching both channels and retuning the ^{1}H channel for ^{19}F. I acquired ^{13}C CP-MAS spectra with variable contact times, ^{19}F MAS spectra, variable-temperature ^{2}H spectra and some ^{15}N CP-MAS spectra. I simulated ^{2}H lineshapes with NMR WebLab to obtain rotational rates and activation energies, and analysed ^{13}C T_1 and ^{1}H and ^{19}F relaxometry data to follow faster motions.

I use NMR together with diffraction. In a pyrrolo[3,2-b]pyrrole solvate that jumps on heating (Mater. Chem. Front., 2026), ^{2}H NMR and variable-temperature single-crystal diffraction allowed me to separate the molecular dynamics from the anisotropic thermal expansion that drives the jump. In a zirconium MOF (ACS Mater. Lett., 2026), the same combination showed how a photoswitch and guest uptake change the motion of a molecular rotor. Both projects were done with Dr Martin Dračínský and Dr Jan Blahut at IOCB Prague.

The role also involves preparing the solid solutions, and synthesis and crystallisation are a large part of my training. I grow single crystals of molecular crystals, cocrystals and solvates and characterise them by SCXRD, PXRD, DSC, TGA and hot-stage microscopy. I contributed to a study in which the salient temperature of cocrystals was tuned through the composition of the crystallisation solvent mixture (Angew. Chem. Int. Ed., 2026). I also run DFT (Gaussian16, periodic CP2K) and molecular dynamics (LAMMPS) on HPC clusters. Diffraction gives the average structure of a molecular solid solution, while solid-state NMR can show how a dopant is distributed and how it moves locally, and this is the question I would like to work on within MoSS.

I have not yet used DNP. The UK DNP MAS NMR Facility at Manchester, and the chance to extend my work to ^{17}O and natural-abundance ^{15}N, are among my main reasons for applying. During my PhD I learned periodic DFT and molecular dynamics from scratch and now use both routinely, and I would approach DNP in the same way.

My 14 publications include collaborations with groups in the Czech Republic, the United States, China and the United Arab Emirates. I co-supervised four bachelor's theses and nine undergraduate research placements, trained students in laboratory procedures and safety, and handled purchasing and inventory for our laboratory. I have presented 15 posters and 9 talks, and received the Best Poster Award at DynaMIC23 in Fribourg. Thank you for considering my application; I would be glad to discuss it at interview.

Yours sincerely,

Dazaet Galicia-Badillo, PhD



Dazaet Galicia-Badillo

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About me

My research has been focused on the design and synthesis of organic and metal-organic framework materials exhibiting dynamics across multiple length scales, from molecular-level rotational motion and photoisomerization to macroscopic phenomena such as crystal flexibility, salient mechanical responses, and single-crystal-to-single-crystal phase transitions. I bring a strong experimental toolkit: organic synthesis, multinuclear solid-state NMR (^{13}C, ^{19}F, ^{2}H, ^{15}N), single-crystal and powder X-ray diffraction (SCXRD/PXRD), fluorimetry, and DFT calculations. My long-term goal is to build an independent research program that bridges materials chemistry with applied challenges in energy storage, environmental remediation, or biomedical applications in academia or the private sector.

Honors, Awards & Funding

Nominated* Alfonso Caso Medal nomination - UNAM. 2026
Nominated for recognition of outstanding postgraduate achievement (*outcome pending).
Ph.D. awarded with honors Chemical Sciences - UNAM.
2026
Best Presentation Award - IX Seminario Regional de Materiales Avanzados, UAEH, Mexico, October
2025
Supercomputing resource allocations – National High-Performance Computing Laboratory Search (LANCAD)
2024
Prepared successful proposals securing two allocations of computational resources for research.
Best Poster Award - DynaMIC23, Fribourg, Switzerland
2023
Dynamic Materials, Crystals and Phenomena Conference hosted by Prof. Simon Krause
Doctoral scholarship - Secretariat of Science, Humanities, Technology, and Innovation (SECIHTI) 2022
Support for Ph.D. studies
Master's scholarship - Secretariat of Science, Humanities, Technology, and Innovation (SECIHTI) 2020
Support for Master's studies
Institute of Chemistry Undergraduate Scholarship -UNAM
2019
Awarded for the best undergraduate research project at the Institute of Chemistry

Education

PhD Chemical Sciences - UNAM, Institute of Chemistry 2022 - 2026
Thesis: Molecular and Macroscopic Motion in Organic Crystals and Metal-Organic Frameworks
Advisor: Prof. Dr. Braulio Rodríguez-Molina
M.Sc. in Chemical Sciences - UNAM, Institute of Chemistry 2020 - 2022
B.S. in Chemistry - UNAM, Faculty of Chemistry 2015 - 2020

Technical skills

Experimental: Organic synthesis, including air- and moisture-sensitive reactions (Schlenk techniques); MOF solvothermal synthesis and post-synthetic modification (PSM). Solid-state NMR (Bruker 500 MHz, double-resonance ^{1}H/X MAS probe): ^{13}C CP-MAS with variable contact time, ^{19}F MAS, variable-temperature ^{2}H and ^{15}N CP-MAS; probe tuning and matching; ^{2}H lineshape simulation; analysis of ^{13}C T_1 and solid-state ^{1}H and ^{19}F relaxometry data. X-ray diffraction: SCXRD and PXRD, including variable-temperature studies and structure refinement (ShelX). Photophysics: steady-state and time-resolved fluorescence, UV–Vis absorption. Thermal analysis: hot-stage microscopy, DSC, TGA. **Computational & Software:** DFT calculations (geometry optimization, electronic structure, energy profiles, PES) in Gaussian16; periodic DFT in CP2K and molecular dynamics simulations (LAMMPS) on HPC clusters. Crystallography tools: ShelX, CrystalDiffract, CCDC suite. Visualization: CrystalMaker, ChemDraw. NMR processing and simulation: TopSpin, MestreNova, NMR WebLab. AI-assisted workflows for research automation and data analysis. **Lab management:** Purchasing

and vendor management; inventory control and lab organization; team leadership and training; funding acquisition (grants/proposals).

PUBLICATIONS (RESEARCH, REVIEW AND PREVIEW)

14. Ernesto Enriquez-Palacios, **Dazaet Galicia-Badillo**, Marcos Flores-Alamo, Jose Luis Belmonte-Vazquez. Imidazole-based Photoswitches by Design and their use for Data Encryption. *ACS Applied Materials & Interfaces*. **2026.** 18, 35, 47895–47906. *[IF: 8.5]*

13. Dazaet Galicia-Badillo, Ernesto A. Hernández-Morales, Yesterday López-Herrera, Erick Hernández-Santiago, Jan Blahut, Rubén A. Toscano, Martin Dracínský, Braulio Rodríguez-Molina. Dissecting Molecular and Macroscopic Dynamics in a Jumping Solvate. *Materials Chemistry Frontiers,* **2026**. 10, 18, 3305-3313. *[IF: 8.1]*

12. Dazaet Galicia-Badillo, Jade Cisneros, Jan Blahut, Alonso Acosta-Vera, Alfonso García-Márquez, Haggeo Desirena, Erika Bartůnková, Ma. Carmen García-González, Ingrid Chávez-Serrano, Joelis Rodríguez-Hernández, Martin Dracínský, Braulio Rodríguez-Molina. Integration of a Molecular Rotor and a Photoswitch in a Metal-Organic Framework. *ACS Materials Letters*, **2026**. 8, 7, 1995-2001. *[IF: 8.7]*

11. Ernesto A. Hernández-Morales, Dayra Barreto-Hernández, **Dazaet Galicia-Badillo**, Rubén A. Toscano, M. Elena García-Aguilera, Braulio Rodríguez-Molina. Crystals as Rockets: Modulation of the Salient Temperature in Cocrystals by Solvent Mixture Composition. *Angewandte Chemie Int. Ed.*, **2026**, e26010. *[IF: 16.9]*

10. Ángel D. Álvarez-Castillo, Alexis Juárez-Morales, **Dazaet Galicia-Badillo**, Mauricio Maldonado-Domínguez, José M. Heredia-Peñaloza, Marcos Flores-Álamo, Jose Luis Belmonte-Vázquez. Rational Modulation of Excited-State Intramolecular Proton Transfer, Aggregation-Induced Emission, and Intramolecular Motion on Polysubstituted Imidazoles (Front Cover). *ChemPhotoChem*, **2026**, 10, e70099. *[IF: 3.0]*

9. Mario Castañón-García, Pedro López-Mendoza, **Dazaet Galicia-Badillo**, Luis D. Miranda. Synthesis of Non-Symmetrical 3,3'-Bicoumarins: Total Synthesis of Arteminorin C, 3,3'-Biisofraxidin and Biscopoletin. *Organic Letters*, **2025**, 27 (21), 5349-5354. *[IF: 5.0]*

8. José L. Belmonte-Vázquez, Durga Prasad Karothu, Carl H. Fleischer, **Dazaet Galicia-Badillo**, Mauricio Maldonado-Domínguez, Robert W. Schurko, Liang Li, Pance Naumov, Braulio Rodríguez-Molina. Low-Temperature Dynamics at Nano- and Macroscales: Organic Crystal That Exhibits Low-Temperature Molecular Motion and the Thermosalient Effect. *Chemistry of Materials*, **2025**, 37, 3373-3383. *[IF: 7.2]*

7. Dazaet Galicia-Badillo, Braulio Rodríguez-Molina. On the Flexibility of 1,4-Bis(phenylethynyl)benzene Butyl Ester Organic Crystal Featuring Mobile Components. *Tetrahedron*, **2024**, 167, 134245. *[IF: 2.2]*

6. Dazaet Galicia-Badillo, José L. Belmonte-Vázquez, Mario Rodríguez, Braulio Rodríguez-Molina, Ma. Carmen García-González. Aggregation-Induced Emission Enhancement and Solid-State Photoswitching of Crystalline Carbazole N-Salicylidene Anilines. *ACS Omega*, **2024**, 9 (36), 38015-38022. *[IF: 4.3]*

5. Alonso Acosta-Vera, **Dazaet Galicia-Badillo**, Armando Navarro-Huerta, Konstantin Stracke, Virginia Gómez-Vidales, Joelis Rodríguez-Hernández, Jack D. Evans, Braulio Rodríguez-Molina. Modulation of the Photophysics and Internal Dynamics in a Zr Metal-Organic Framework by the Inclusion of Fluorinated Guests. *ACS Materials Letters*, **2024**, 6 (9), 4395-4401. *[IF: 8.7]*

4. Dazaet Galicia-Badillo, Paulina Noemí García-Vargas, Braulio Rodríguez-Molina. Preview: The Integration of Multiple Components for Multi-Responsive Dynamic Framework Materials. *Matter*, **2024**, 7 (8), 2766-2768. *[IF: 17.5]*

3. Erick Hernández-Santiago, Jorge Espinosa-Rocha, **Dazaet Galicia-Badillo**, Braulio Rodríguez-Molina. Review: Local and Global Dynamics in Organic Soft Porous Crystals. *Matter*, **2024**, 7 (2), 408-429. *[IF: 17.5]*

2. Manuel Rodríguez-Molina, **Dazaet Galicia-Badillo**, Enoc Cetina-Mancilla, Jorge Cárdenas, Lilian I. Olvera, Rubén A. Toscano, Braulio Rodríguez-Molina, Mikhail G. Zolotukhin. 9-Trifluoromethylxanthenediols: Synthesis and Supramolecular Motifs. *ACS Omega*, **2022**, 7 (16), 13520-13528. *[IF: 4.1]*

1. Aaron Torres-Huerta, **Dazaet Galicia-Badillo**, Andrés Aguilar-Granda, Jacob T. Bryant, Fernando J. Uribe-Romo, Braulio Rodríguez-Molina. Multiple Rotational Rates in a Guest-Loaded, Amphidynamic Zirconia Metal-Organic Framework. *Chemical Science*, **2020**, 11 (42), 11579-11583. *[IF: 9.8]*

TEACHING & MENTORSHIP

Co-director of 4 bachelor's theses (2023-present), covering MOF synthesis, organic crystal dynamics, and luminescent materials. Supervisor of 9 undergraduate short research stays (2022-2025) and 1 high school student.

Scientific conferences & presentations

15 posters (6 international, 9 national) · 9 oral presentations (4 invited lectures, 5 congress talks)

Scientific divulgation

1 article (Boletín IQ-UNAM, 2022) · 2 public lectures (High Schools, 2024–2025) · 1 media consultation (UNAM Global, 2022)

References

Prof. Dr. Braulio Rodríguez Molina - PhD Supervisor**.** Instituto de Química, Universidad Nacional Autónoma de México
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